

In Problems 1–16, use separation of variables to find, if possible, product solutions for the given partial differential equation.

1. $\frac{\partial u}{\partial x} = \frac{\partial u}{\partial y}$
2. $\frac{\partial u}{\partial x} + 3 \frac{\partial u}{\partial y} = 0$
3. $u_x + u_y = u$
4. $u_x = u_y + u$
5. $x \frac{\partial u}{\partial x} = y \frac{\partial u}{\partial y}$
6. $y \frac{\partial u}{\partial x} + x \frac{\partial u}{\partial y} = 0$
7. $\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial x \partial y} + \frac{\partial^2 u}{\partial y^2} = 0$
8. $y \frac{\partial^2 u}{\partial x \partial y} + u = 0$
9. $k \frac{\partial^2 u}{\partial x^2} - u = \frac{\partial u}{\partial t}, k > 0$
10. $k \frac{\partial^2 u}{\partial x^2} = \frac{\partial u}{\partial t}, k > 0$
11. $a^2 \frac{\partial^2 u}{\partial x^2} = \frac{\partial^2 u}{\partial t^2}$
12. $a^2 \frac{\partial^2 u}{\partial x^2} = \frac{\partial^2 u}{\partial t^2} + 2k \frac{\partial u}{\partial t}, k > 0$
13. $\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} + 2k \frac{\partial u}{\partial t}, k > 0$
14. $x^2 \frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} = 0$
15. $u_{xx} + u_{yy} = u$
16. $a^2 u_{xx} - g = u_t, g \text{ a constant}$

In Problems 17–26, classify the given partial differential equation as hyperbolic, parabolic, or elliptic.

17. $\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial x \partial y} + \frac{\partial^2 u}{\partial y^2} = 0$
18. $3 \frac{\partial^2 u}{\partial x^2} + 5 \frac{\partial^2 u}{\partial x \partial y} + \frac{\partial^2 u}{\partial y^2} = 0$
19. $\frac{\partial^2 u}{\partial x^2} + 6 \frac{\partial^2 u}{\partial x \partial y} + 9 \frac{\partial^2 u}{\partial y^2} = 0$
20. $\frac{\partial^2 u}{\partial x^2} - \frac{\partial^2 u}{\partial x \partial y} - 3 \frac{\partial^2 u}{\partial y^2} = 0$
21. $\frac{\partial^2 u}{\partial x^2} = 9 \frac{\partial^2 u}{\partial x \partial y}$
22. $\frac{\partial^2 u}{\partial x \partial y} - \frac{\partial^2 u}{\partial y^2} + 2 \frac{\partial u}{\partial x} = 0$

$$23. \frac{\partial^2 u}{\partial x^2} + 2 \frac{\partial^2 u}{\partial x \partial y} + \frac{\partial^2 u}{\partial y^2} + \frac{\partial u}{\partial x} - 6 \frac{\partial u}{\partial y} = 0$$

$$24. \frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} = u$$

$$25. a^2 \frac{\partial^2 u}{\partial x^2} = \frac{\partial^2 u}{\partial t^2} \quad 26. k \frac{\partial^2 u}{\partial x^2} = \frac{\partial u}{\partial t}, k > 0$$

In Problems 27 and 28, show that the given partial differential equation possesses the indicated product solution.

$$27. k \left(\frac{\partial^2 u}{\partial r^2} + \frac{1}{r} \frac{\partial u}{\partial r} \right) = \frac{\partial u}{\partial t};$$

$$u = e^{-k\alpha^2 t} (c_1 J_0(\alpha r) + c_2 Y_0(\alpha r))$$

$$28. \frac{\partial^2 u}{\partial r^2} + \frac{1}{r} \frac{\partial u}{\partial r} + \frac{1}{r^2} \frac{\partial^2 u}{\partial \theta^2} = 0;$$

$$u = (c_1 \cos \alpha \theta + c_2 \sin \alpha \theta)(c_3 r^\alpha + c_4 r^{-\alpha})$$

29. Verify that each of the products $u = X(x)Y(y)$ in (6), (7), and (8) satisfies the second-order PDE in Example 2.

30. Definition 13.1.1 generalizes to linear PDEs with coefficients that are functions of x and y . Determine the regions in the xy -plane for which the equation

$$(xy + 1) \frac{\partial^2 u}{\partial x^2} + (x + 2y) \frac{\partial^2 u}{\partial x \partial y} + \frac{\partial^2 u}{\partial y^2} + xy^2 u = 0$$

is hyperbolic, parabolic, or elliptic.

Discussion Problems

In Problems 31 and 32, discuss whether product solutions $u = X(x)Y(y)$ can be found for the given partial differential equation. [Hint: Use the superposition principle.]

$$31. \frac{\partial^2 u}{\partial x^2} - u = 0$$

$$32. \frac{\partial^2 u}{\partial x \partial y} + \frac{\partial u}{\partial x} = 0$$

13.2

Classical PDEs and Boundary-Value Problems

INTRODUCTION For the remainder of this and the next chapter we shall be concerned with finding product solutions of the second-order partial differential equations

$$k \frac{\partial^2 u}{\partial x^2} = \frac{\partial u}{\partial t}, \quad k > 0 \quad (1)$$

$$a^2 \frac{\partial^2 u}{\partial x^2} = \frac{\partial^2 u}{\partial t^2} \quad (2)$$

$$\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} = 0 \quad (3)$$

or slight variations of these equations. These classical equations of mathematical physics are known, respectively, as the **one-dimensional heat equation**, the **one-dimensional wave equation**, and **Laplace's equation in two dimensions**. “One-dimensional” refers to the fact that x denotes a spatial dimension whereas t represents time; “two dimensional” in (3) means that x and y are both spatial dimensions. Laplace's equation is abbreviated $\nabla^2 u = 0$, where

$$\nabla^2 u = \frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2}$$

is called the **two-dimensional Laplacian** of the function u . In three dimensions the **Laplacian** of u is

$$\nabla^2 u = \frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} + \frac{\partial^2 u}{\partial z^2}.$$

By comparing equations (1)–(3) with the linear second-order PDE given in Definition 13.1.1, with t playing the part of y , we see that the heat equation (1) is parabolic, the wave equation (2) is hyperbolic, and Laplace's equation (3) is elliptic. This classification is important in Chapter 16.

Heat Equation Equation (1) occurs in the theory of heat flow—that is, heat transferred by conduction in a rod or thin wire. The function $u(x, t)$ is temperature. Problems in mechanical vibrations often lead to the wave equation (2). For purposes of discussion, a solution $u(x, t)$ of (2) will represent the displacement of an idealized string. Finally, a solution $u(x, y)$ of Laplace's equation (3) can be interpreted as the steady-state (that is, time-independent) temperature distribution throughout a thin, two-dimensional plate.

Even though we have to make many simplifying assumptions, it is worthwhile to see how equations such as (1) and (2) arise.

Suppose a thin circular rod of length L has a cross-sectional area A and coincides with the x -axis on the interval $[0, L]$. See **FIGURE 13.2.1**. Let us suppose:

- The flow of heat within the rod takes place only in the x -direction.
- The lateral, or curved, surface of the rod is insulated; that is, no heat escapes from this surface.
- No heat is being generated within the rod by either chemical or electrical means.
- The rod is homogeneous; that is, its mass per unit volume ρ is a constant.
- The specific heat γ and thermal conductivity K of the material of the rod are constants.

To derive the partial differential equation satisfied by the temperature $u(x, t)$, we need two empirical laws of heat conduction:

- (i) *The quantity of heat Q in an element of mass m is*

$$Q = \gamma m u, \quad (4)$$

where u is the temperature of the element.

- (ii) *The rate of heat flow Q_t through the cross section indicated in Figure 13.2.1 is proportional to the area A of the cross section and the partial derivative with respect to x of the temperature:*

$$Q_t = -K A u_x. \quad (5)$$

Since heat flows in the direction of decreasing temperature, the minus sign in (5) is used to ensure that Q_t is positive for $u_x < 0$ (heat flow to the right) and negative for $u_x > 0$ (heat flow to the left). If the circular slice of the rod shown in Figure 13.2.1 between x and $x + \Delta x$ is very thin, then $u(x, t)$ can be taken as the approximate temperature at each point in the interval. Now the mass of the slice is $m = \rho(A \Delta x)$, and so it follows from (4) that the quantity of heat in it is

$$Q = \gamma \rho A \Delta x u. \quad (6)$$

Furthermore, when heat flows in the positive x -direction, we see from (5) that heat builds up in the slice at the net rate

$$-K A u_x(x, t) - [-K A u_x(x + \Delta x, t)] = K A [u_x(x + \Delta x, t) - u_x(x, t)]. \quad (7)$$

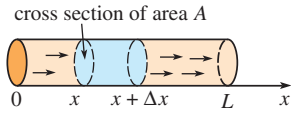


FIGURE 13.2.1 One-dimensional flow of heat

By differentiating (6) with respect to t we see that this net rate is also given by

$$Q_t = \gamma \rho A \Delta x u_t. \quad (8)$$

Equating (7) and (8) gives

$$\frac{K}{\gamma \rho} \frac{u_x(x + \Delta x, t) - u_x(x, t)}{\Delta x} = u_t. \quad (9)$$

Taking the limit of (9) as $\Delta x \rightarrow 0$ finally yields (1) in the form*

$$\frac{K}{\gamma \rho} u_{xx} = u_t.$$

It is customary to let $k = K/\gamma \rho$ and call this positive constant the **thermal diffusivity**.

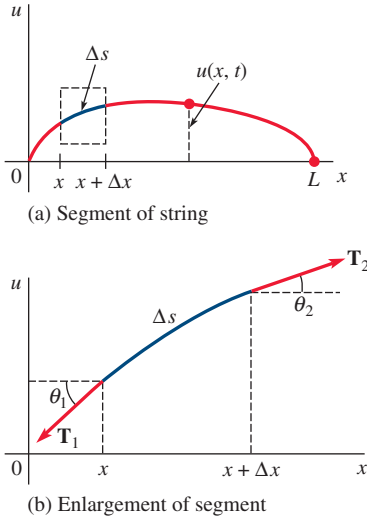


FIGURE 13.2.2 Taut string anchored at two points on the x -axis

Wave Equation Consider a string of length L , such as a guitar string, stretched taut between two points on the x -axis—say, $x = 0$ and $x = L$. When the string starts to vibrate, assume that the motion takes place in the xy -plane in such a manner that each point on the string moves in a direction perpendicular to the x -axis (transverse vibrations). As shown in **FIGURE 13.2.2(a)**, let $u(x, t)$ denote the vertical displacement of any point on the string measured from the x -axis for $t > 0$. We further assume:

- The string is perfectly flexible.
- The string is homogeneous; that is, its mass per unit length ρ is a constant.
- The displacements u are small compared to the length of the string.
- The slope of the curve is small at all points.
- The tension $\mathbf{T}$ acts tangent to the string, and its magnitude T is the same at all points.
- The tension is large compared with the force of gravity.
- No other external forces act on the string.

Now in **Figure 13.2.2(b)** the tensions $\mathbf{T}_1$ and $\mathbf{T}_2$ are tangent to the ends of the curve on the interval $[x, x + \Delta x]$. For small values of θ_1 and θ_2 the net vertical force acting on the corresponding element Δs of the string is then

$$\begin{aligned} T \sin \theta_2 - T \sin \theta_1 &\approx T \tan \theta_2 - T \tan \theta_1 \\ &= T[u_x(x + \Delta x, t) - u_x(x, t)],^\dagger \end{aligned}$$

where $T = |\mathbf{T}_1| = |\mathbf{T}_2|$. Now $\rho \Delta s \approx \rho \Delta x$ is the mass of the string on $[x, x + \Delta x]$, and so Newton's second law gives

$$T[u_x(x + \Delta x, t) - u_x(x, t)] = \rho \Delta x u_{tt}$$

or

$$\frac{u_x(x + \Delta x, t) - u_x(x, t)}{\Delta x} = \frac{\rho}{T} u_{tt}.$$

If the limit is taken as $\Delta x \rightarrow 0$, the last equation becomes $u_{xx} = (\rho/T)u_{tt}$. This of course is (2) with $a^2 = T/\rho$.

Laplace's Equation Although we shall not present its derivation, Laplace's equation in two and three dimensions occurs in time-independent problems involving potentials such as electrostatic, gravitational, and velocity in fluid mechanics. Moreover, a solution of Laplace's equation can also be interpreted as a steady-state temperature distribution. As illustrated in **FIGURE 13.2.3**, a solution $u(x, y)$ of (3) could represent the temperature that varies from point to point—but not with time—of a rectangular plate.

We often wish to find solutions of equations (1), (2), and (3) that satisfy certain side conditions.

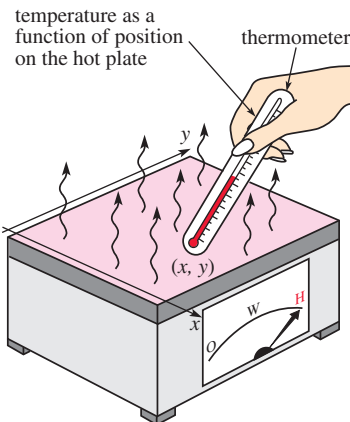


FIGURE 13.2.3 Steady-state temperatures in a rectangular plate

*Recall from calculus that $u_{xx} = \lim_{\Delta x \rightarrow 0} \frac{u_x(x + \Delta x, t) - u_x(x, t)}{\Delta x}$.

$^\dagger \tan \theta_2 = u_x(x + \Delta x, t)$ and $\tan \theta_1 = u_x(x, t)$ are equivalent expressions for slope.

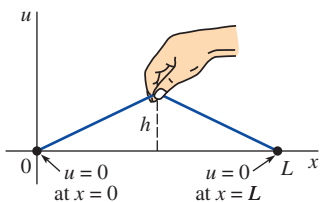


FIGURE 13.2.4 Plucked string

Initial Conditions Since solutions of (1) and (2) depend on time t , we can prescribe what happens at $t = 0$; that is, we can give **initial conditions (IC)**. If $f(x)$ denotes the initial temperature distribution throughout the rod in Figure 13.2.1, then a solution $u(x, t)$ of (1) must satisfy the single initial condition $u(x, 0) = f(x)$, $0 < x < L$. On the other hand, for a vibrating string, we can specify its initial displacement (or shape) $f(x)$ as well as its initial velocity $g(x)$. In mathematical terms we seek a function $u(x, t)$ satisfying (2) and the two initial conditions:

$$u(x, 0) = f(x), \quad \left. \frac{\partial u}{\partial t} \right|_{t=0} = g(x), \quad 0 < x < L. \quad (10)$$

For example, the string could be plucked, as shown in **FIGURE 13.2.4**, and released from rest ($g(x) = 0$).

Boundary Conditions The string in Figure 13.2.4 is secured to the x -axis at $x = 0$ and $x = L$ for all time. We interpret this by the two **boundary conditions (BC)**:

$$u(0, t) = 0, \quad u(L, t) = 0, \quad t > 0.$$

Note that in this context the function f in (10) is continuous, and consequently $f(0) = 0$ and $f(L) = 0$. In general, there are three types of boundary conditions associated with equations (1), (2), and (3). On a boundary we can specify the values of *one* of the following:

$$(i) \quad u, \quad (ii) \quad \frac{\partial u}{\partial n}, \quad \text{or} \quad (iii) \quad \frac{\partial u}{\partial n} + hu, \quad h \text{ a constant.}$$

Here $\partial u / \partial n$ denotes the normal derivative of u (the directional derivative of u in the direction perpendicular to the boundary). A boundary condition of the first type (i) is called a **Dirichlet condition**, a boundary condition of the second type (ii) is called a **Neumann condition**, and a boundary condition of the third type (iii) is known as a **Robin condition**. For example, for $t > 0$ a typical condition at the right-hand end of the rod in Figure 13.2.1 can be

$$(i)' \quad u(L, t) = u_0, \quad u_0 \text{ a constant,}$$

$$(ii)' \quad \left. \frac{\partial u}{\partial x} \right|_{x=L} = 0, \quad \text{or}$$

$$(iii)' \quad \left. \frac{\partial u}{\partial x} \right|_{x=L} = -h(u(L, t) - u_m), \quad h > 0 \text{ and } u_m \text{ constants.}$$

Condition (i)' simply states that the boundary $x = L$ is held by some means at a constant *temperature* u_0 for all time $t > 0$. Condition (ii)' indicates that the boundary $x = L$ is *insulated*. From the empirical law of heat transfer, the flux of heat across a boundary (that is, the amount of heat per unit area per unit time conducted across the boundary) is proportional to the value of the normal derivative $\partial u / \partial n$ of the temperature u . Thus when the boundary $x = L$ is thermally insulated, no heat flows into or out of the rod and so

$$\left. \frac{\partial u}{\partial x} \right|_{x=L} = 0.$$

We can interpret (iii)' to mean that *heat is lost* from the right-hand end of the rod by being in contact with a medium, such as air or water, that is held at a constant temperature. From Newton's law of cooling, the outward flux of heat from the rod is proportional to the difference between the temperature $u(L, t)$ at the boundary and the temperature u_m of the surrounding medium. We note that if heat is lost from the left-hand end of the rod, the boundary condition is

$$\left. \frac{\partial u}{\partial x} \right|_{x=0} = h(u(0, t) - u_m).$$

The change in algebraic sign is consistent with the assumption that the rod is at a higher temperature than the medium surrounding the ends so that $u(0, t) > u_m$ and $u(L, t) > u_m$. At $x = 0$ and $x = L$, the slopes $u_x(0, t)$ and $u_x(L, t)$ must be positive and negative, respectively.

Of course, at the ends of the rod we can specify different conditions at the same time. For example, we could have

$$\left. \frac{\partial u}{\partial x} \right|_{x=0} = 0 \quad \text{and} \quad u(L, t) = u_0, \quad t > 0.$$

We note that the boundary condition in (i)' is homogeneous if $u_0 = 0$; if $u_0 \neq 0$, the boundary condition is nonhomogeneous. The boundary condition (ii)' is homogeneous; (iii)' is homogeneous if $u_m = 0$ and nonhomogeneous if $u_m \neq 0$.

Boundary-Value Problems Problems such as

$$\begin{aligned} \text{Solve:} \quad & a^2 \frac{\partial^2 u}{\partial x^2} = \frac{\partial^2 u}{\partial t^2}, \quad 0 < x < L, \quad t > 0 \\ \text{Subject to:} \quad & \text{(BC)} \quad u(0, t) = 0, \quad u(L, t) = 0, \quad t > 0 \\ & \text{(IC)} \quad u(x, 0) = f(x), \quad \left. \frac{\partial u}{\partial t} \right|_{t=0} = g(x), \quad 0 < x < L \end{aligned} \quad (11)$$

and

$$\begin{aligned} \text{Solve:} \quad & \frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} = 0, \quad 0 < x < a, \quad 0 < y < b \\ \text{Subject to:} \quad & \text{(BC)} \quad \begin{cases} \left. \frac{\partial u}{\partial x} \right|_{x=0} = 0, \quad \left. \frac{\partial u}{\partial x} \right|_{x=a} = 0, \quad 0 < y < b \\ u(x, 0) = 0, \quad u(x, b) = f(x), \quad 0 < x < a \end{cases} \end{aligned} \quad (12)$$

are called **boundary-value problems**. The problem in (11) is classified as a **homogeneous BVP** since the partial differential equation and the boundary conditions are homogeneous.

Variations The partial differential equations (1), (2), and (3) must be modified to take into consideration internal or external influences acting on the physical system. More general forms of the one-dimensional heat and wave equations are, respectively,

$$k \frac{\partial^2 u}{\partial x^2} + F(x, t, u, u_x) = \frac{\partial u}{\partial t} \quad (13)$$

and

$$a^2 \frac{\partial^2 u}{\partial x^2} + F(x, t, u, u_t) = \frac{\partial^2 u}{\partial t^2}. \quad (14)$$

For example, if there is heat transfer from the lateral surface of a rod into a surrounding medium that is held at a constant temperature u_m , then the heat equation (13) is

$$k \frac{\partial^2 u}{\partial x^2} - h(u - u_m) = \frac{\partial u}{\partial t},$$

where h is a constant. In (14) the function F could represent the various forces acting on the string. For example, when external, damping, and elastic restoring forces are taken into account, (14) assumes the form

$$\begin{array}{ccccc} \text{external force} & & \text{damping} & & \text{restoring force} \\ & \downarrow & \downarrow & \downarrow & \\ a^2 \frac{\partial^2 u}{\partial x^2} + \underbrace{f(x, t) - c \frac{\partial u}{\partial t} - ku}_{F(x, t, u, u_t)} = \frac{\partial^2 u}{\partial t^2}. \end{array} \quad (15)$$